Digital Banking — System Architecture Overview

Zava Digital Operations · Architecture Reference Document · Confidential

1. Edge Layer

CloudFront / Front Door edge → WAF → API Gateway. Region routing per geo. Static assets cached at edge.

2. API Gateway Layer

API Gateway (regional). Rate limiting, JWT validation, request routing to downstream services. Tied to upstream timeouts.

3. Application Services

Auth Service (stateful, Redis-backed session store, OIDC provider integration). Transaction Service (stateful, calls into core banking). Notification Service (Kafka-fed). Customer Profile Service. Reporting Service.

4. Data Tier

Primary OLTP — Postgres clusters (regional). Session store — Redis (regional with cross-region async replication). Core banking — mainframe (system-of-record).

5. Messaging

Kafka clusters for event streaming (notifications, audit log, fraud signals).

6. Observability

Metrics: Prometheus → Grafana. Logs: ELK. Traces: OpenTelemetry. Synthetic monitoring + RUM (real user monitoring).

7. Critical Dependencies

Auth Service ← Session Store ← OIDC Provider (3rd-party). Transaction Service ← Core Banking. Notification Service ← SMS gateway (3rd-party).

8. Failure Domain Notes

Session Store is the historical bottleneck. Connection pool saturation has been observed in 2023 and 2024 under similar load patterns. Auto-scaling triggers exist but lag by 90 seconds — within that window, downstream auth queue saturates rapidly.

9. Recent Changes (Risk-Relevant, last 30 days)

2026-04-08 — connection pool default sizing changed from 200 to 80 in a config refactor (intended for non-prod default; mistakenly inherited by prod via the release pipeline) — root cause of the 2026-04-18 incident.

10. Resilience Posture

Multi-region (4 regions). Active-Active for read traffic; Active-Passive for writes. RTO target 15 min; RPO 1 minute.